

**HIMACHAL PRADESH TECHNICAL UNIVERSITY
HAMIRPUR**



Syllabus
for
MCA (Master of Computer Applications)

As per National Education Policy (NEP-2020)

(w.e.f. the Academic Year 2023-2024)

Department of Master of Computer Applications
School of Computer Science and Engineering

Approved by the Board of Studies

Scheme of Teaching and Examination Master of Computer Applications (MCA)

Semester-III

Subject Code	Course Category	Subject Title/ Subject Name	Periods			Credits	Evaluation Scheme					Total
			L	T	P		ESE	Internal Assessment				
								PE	TA	A	Total	
MCA-7301	GE	Operational Research	3	0	0	3	60	20	15	05	40	100
MCA-7302	CC	Analysis Design and Algorithm	3	0	0	3	60	20	15	05	40	100
MCA-7303	CC	Cloud Computing	3	0	0	3	60	20	15	05	40	100
MCA-7304	SEC	Mobile Application Deployment	3	0	0	3	60	20	15	05	40	100
MCA-7305	DSE	DSE - II	4	0	0	4	60	20	15	05	40	100
MCA-7306	DSE	DSE - III	4	0	0	4	60	20	15	05	40	100
Labs												
MCA-7307P	CC LAB	Lab VII: Analysis Design and Algorithm Lab	0	0	4	2	60	20	15	05	40	100
MCA-7308P	SEC LAB	Lab VIII:Mobile Application Deployment Lab	0	0	4	2	60	20	15	05	40	100
MCA-7309P	CC LAB	Lab IX: Innovative Project Identification	0	0	4	2	60	20	15	05	40	100
Total			20	0	12	26	540	180	135	45	360	900

Legends:	CC - Core Course	ESE-End Semester Examination
	SEC - Skill Enhancement Course	PE – Periodical Examination
	VAC – Value Addition Course	TA - Teacher's Assessment
	DSE - Discipline Specific Electives	A – Attendance
	GE – Generic Elective	L – Lecture
	IDE - Inter Departmental Elective Course	T – Tutorial
	C – Compulsory	P – Practical

Departmental Electives University Wide Courses

Discipline Specific Electives- II

Sr. No.	Course Type	Subject Code	Subject Title/ Subject Name
1.	DSE	MCA- 7305(i)	Compiler Design
2.	DSE	MCA- 7305(ii)	Data Warehouse and Mining
3.	DSE	MCA- 7305(iii)	Ethical Hacking
4.	DSE	MCA- 7305(iv)	Machine Learning
5.	DSE	MCA- 7305 (v)	Introduction to Quantum Computing

Discipline Specific Electives- III

Sr. No.	Course Type	Subject Code	Subject Title/ Subject Name
1.	DSE	MCA- 7306(i)	Internet of Things (IoT)
2.	DSE	MCA- 7306(ii)	Soft Computing
3.	DSE	MCA- 7306(iii)	Computer Vision
4.	DSE	MCA- 7306(iv)	Introduction to Blockchain
5.	DSE	MCA- 7306(v)	Software Engineering

Semester-IV

Course Code	Course Type	Subject Title/ Subject Name	Credits	Marks	
MCA-7401(A)	CC	Innovative Project Design & Implementation (Industrial / In-House)	16	Min	Max
		(A). Synopsis		20	50
		(B). Innovative Mid Project Report		20	50
		(C). Innovative Project Seminars(02)		20	50
		(D). Innovative Project Progress Report(02)		20	50
		(E). Innovative Project Report		40	100
		(F). Major Project Viva-Voce		40	100
Total				160	400
Or					
MCA-7401(B)	CC	Research (Industrial / In-House)	16	Marks	
				Min	Max
		(A). Synopsis		20	50
		(B). Mid Research Report		20	50
		(C). Research Work Seminars (02)		20	50
		(D). Research Work Progress Report(02)		20	50
		(E). Research Work Report		40	100
		(F). Research Work Viva-Voce		40	100
		Total	160	400	

HIMACHAL PRADESH TECHNICAL UNIVERSITY HAMIRPUR



Syllabus

for

MCA

(SEMESTER-III)

Amended as per NEP-2020

(w.e.f. the Academic Year 2023-2024)

MCA-7301 Operational Research							
Teaching Scheme			Credit	Marks Distribution			Duration of End Semester Examination
L	T	P	C	Internal Assessment	End Semester Examination	Total	
3	0	0	3	Maximum Marks: 40	Maximum Marks: 60	100	3 Hours
				Minimum Marks: 16	Minimum Marks: 24	40	

Guidelines for setting Question Paper: Question paper of end semester examination will be of 60 marks. The question paper will consist of five sections A, B, C, D, E. Sections A,B,C and D will have 2 questions of 12 marks each and Section E has short answer type questions consisting of six parts of 02 marks each. The candidates will attempt five questions in all, i.e. one question each from sections A,B,C,D and the compulsory question from section E. In the question paper, the questions available in sections A,B,C and D will be covered from Unit-I, Unit-II, Unit-III and Unit-IV respectively and Section-E will covered the whole syllabus.

Course Objectives (COs)

- This course explains various optimization problems and techniques to address those problems.
- To understand game theory and Queueing theory.

Unit I	08 Lectures
Linear programming problems: Introduction, Uses and Limitations ,Mathematical formulation, graphical method of solution and some exceptional cases,Canonical and standard form of LPP,simplex method, Artificial variable techniques (Big M), Two Phase of Simplex Method, Problem of Degeneracy.	
Unit II	08 Lectures
The Dual Simplex Method: Dual and Primal Problem, Duality and Simplex Method , Transportation and Assignment problems, Traveling salesman Problem. CPM & PERT: Project scheduling, Critical path calculations, Crashing.	
Unit III	07 Lectures
Game Theory: Introduction, two-person zero-sum games, some basic terms, the maxmini minimax principle, games without saddle points-Mixed Strategies, graphic solution of $2 \times n$ and $m \times 2$ games, dominance property. Queueing theory: Basic structure of queuing systems, roles of the Poisson and exponential distributions, classification of queues basic results of M/M/1: FIFO systems, extension to multi-server queues.	
Unit IV	07 Lectures
Inventory Control: Introduction, Inventory Control, Selective Control Techniques, ABC analysis Procedure, Economics lot size problems, Problem EOQ with shortage, Inventory Control Techniques uncertain demand, Stochastic Problems.	

Course Learning Outcomes (CLOs)

- The students will be able to gain the knowledge of optimization techniques.
- Get the skill to apply optimization techniques to address the real time problems.

Suggested Readings

- Taha.H.A ,Operation Research : An Introduction, McMilan publishing Co., 1982. 7th ed
- Ravindran A, Philips D.T & Solbery.J.J, Operations Research: Principles and practice, John Wiley & Sons, New York, 1987.
- Frank S. Budnick, Dennis Mcleavey and Richard Mojena, Principles of Operations Research for Management. All India Traveler Book seller, Delhi.

MCA-7302 Analysis and Design of Algorithm							
Teaching Scheme			Credit	Marks Distribution			Duration of End Semester Examination
L	T	P	C	Internal Assessment	End Semester Examination	Total	
3	0	0	3	Maximum Marks: 40	Maximum Marks: 60	100	3 Hours
				Minimum Marks: 16	Minimum Marks: 24	40	

Guidelines for setting Question Paper: Question paper of end semester examination will be of 60 marks. The question paper will consist of five sections A, B, C, D, E. Sections A,B,C and D will have 2 questions of 12 marks each and Section E has short answer type questions consisting of six parts of 02 marks each. The candidates will attempt five questions in all, i.e. one question each from sections A,B,C, D and the compulsory question from section E. In the question paper, the questions available in sections A,B,C and D will be covered from Unit-I, Unit-II, Unit-III and Unit-IV respectively and Section-E will covered the whole syllabus.

Course Objectives (COs)

The main objective of this course is to describes major algorithmic techniques and mention problems for which each technique is appropriate.

Unit I	08 Lectures
Performance analysis of Algorithm and Recurrences: Time and Space complexities, Asymptotic Notation, Recurrence Relations. Design Techniques: Divide and Conquer, Greedy Algorithms, Backtracking, Branch and Bound.	
Unit II	08 Lectures
Lower Bound Theory: Comparison Tree, Lower Bound through Reductions. Graph Algorithms: Breadth First Search, Depth First Search, Shortest Paths, Maximum Flow, Minimum Spanning Trees.	
Unit III	07 Lectures
Introduction to NP-Completeness: The class P and NP, Polynomial reduction, NP- Completeness Problem, NP-Hard Problems. Advanced Algorithms: Approximation Algorithms, Randomized algorithms.	
Unit IV	07 Lectures
String Matching Algorithm: Introduction, The naive string matching algorithm, The Rabin-Karp algorithm, String Matching with finite automata, The Knuth-Morris-Pratt algorithm.	

Course Learning Outcomes (CLOs)

- The student will be able to understand how the choice of data structures and the algorithm design methods impact the performance of programs.

Suggested Readings:

- Alfred V. Aho, Jeffrey D. Ullman (2001), Principles of compiler design, Indian student edition, Pearson Education, New Delhi, India.

MCA-7303 Cloud Computing							
Teaching Scheme			Credit	Marks Distribution			Duration of End Semester Examination
L	T	P	C	Internal Assessment	End Semester Examination	Total	
3	0	0	3	Maximum Marks: 40	Maximum Marks: 60	100	3 Hours
				Minimum Marks: 16	Minimum Marks: 24	40	

Guidelines for setting Question Paper: Question paper of end semester examination will be of 60 marks. The question paper will consist of five sections A, B, C, D, E. Sections A,B,C and D will have 2 questions of 12 marks each and Section E has short answer type questions consisting of six parts of 02 marks each. The candidates will attempt five questions in all, i.e. one question each from sections A,B,C,D and the compulsory question from section E. In the question paper, the questions available in sections A,B,C and D will be covered from Unit-I, Unit-II, Unit-III and Unit-IV respectively and Section-E will covered the whole syllabus.

Course Objectives (COs)

- The main objective of this course is to provide conceptual as well as practical knowledge of basic of Cloud Computing, Various Cloud Computing terminologies and Platforms. After completing the course the student should be competent in cloud computing concepts and platforms.

Unit I	08 Lectures
Introduction to Cloud Computing: Overview, NIST features, Historical Development, Need for Cloud Computing, Principles of Cloud Computing, Challenges and Risk of Cloud Computing. Advantages and Drawbacks. Cloud Computing Architecture and Reference Model : NIST Cloud Computing Reference Architecture, IBM Cloud Computing Reference Architecture, DMTF Cloud Service Reference Architecture, Cloud applications.	
Unit II	08 Lectures
Cloud Models: Service Model: IaaS, PaaS, SaaS, Other Service Models, Deployment Model: Cloud Deployment Models: Private Cloud, Public Cloud, Community Cloud, Hybrid Cloud Virtualization Technology: Virtualization Structures and Mechanisms, Hypervisor, Type 1 Hypervisor 1 Type 2 Hypervisor, Approaches of Virtualization Types of Virtualizations, Multi-Tenancy, Mobile Computing.	
Unit III	07 Lectures
Cloud Database NoSQL : Relational, Non-Relational vs. DBaaS Cloud Database, Cloud Database Architectures, Cloud Databases, Amazon Dynamo Database, HBase, Cassandra, Google Big Table, Hive, MongoDB. Cloud File System: Google File System (GFS) Vs Hadoop Distributed File System (HDFS)	
Unit IV	07 Lectures
Cloud Security: Cloud Information Security Fundamentals, Security Challenges, Infrastructure Security, Cloud computing security architecture.	

Course Learning Outcomes (CLOs)

- To analyse the problems and challenges associated with building distributed computing applications
- Design high performance and cloud applications to support scalable online services.

Suggested Readings

- Pawan Thakur, Susheela Pathania “Cloud Computing”, Satya Prakashan New Delhi.
- Ronald Krutz and Russell “Cloud Security Dean Vines”, Wiley-India.
- Tim Malhar, S.Kumara Swammy ,“Cloud Security and Privacy” Latif(SPD,O'REILLY).

MCA-7304 Mobile Application Deployment							
Teaching Scheme			Credit	Marks Distribution			Duration of End Semester Examination
L	T	P	C	Internal Assessment	End Semester Examination	Total	
3	0	0	3	Maximum Marks: 40	Maximum Marks: 60	100	3 Hours
				Minimum Marks: 16	Minimum Marks: 24	40	

Guidelines for setting Question Paper: Question paper of end semester examination will be of 60 marks. The question paper will consist of five sections A, B, C, D, E. Sections A,B,C and D will have 2 questions of 12 marks each and Section E has short answer type questions consisting of six parts of 02 marks each. The candidates will attempt five questions in all, i.e. one question each from sections A,B,C, D and the compulsory question from section E. In the question paper, the questions available in sections A,B,C and D will be covered from Unit-I, Unit-II, Unit-III and Unit-IV respectively and Section-E will covered the whole syllabus.

Course Objectives (COs)

- The objectives of this course is to facilitate students to understand android SDK.
- To help students to gain a basic understanding of Android application development.

Unit I	08 Lectures
Introduction to Android: The Android Platform, Android SDK, Eclipse Installation, Android Installation, Building you First Android application, Understanding Anatomy of Android Application, Android Manifest file.	
Unit II	08 Lectures
Android Application Design Essentials: Anatomy of an Android applications, Android terminologies, Application Context, Activities, Services, Intents, Receiving and Broadcasting Intents, Android Manifest File and its common settings, Using Intent Filter, Permissions	
Unit III	07 Lectures
Testing Android applications, Publishing Android application, Using Android preferences, Managing Application resources in a hierarchy, working with different types of resources.	
Unit IV	07 Lectures
Using Common Android APIs: Using Android Data and Storage APIs, Managing data using Sqlite, Sharing Data between Applications with Content Providers, Using Android Networking APIs, Using Android Web APIs, Using Android Telephony APIs, Deploying Android Application to the World.	

Course Learning Outcomes (CLOs)

- The student will be able to identify various concepts of mobile programming that make it unique from programming for other platforms.
- Deploy applications to the Android market place for distribution.

Suggested Readings:

- Lauren Darcey and Shane Conder, “Android Wireless Application Development”, Pearson Education, 2nd ed. (2011).
- Android Application Development All in one for Dummies by Barry Burd, Edition.

MCA-7305(i) Compiler Design							
Teaching Scheme			Credit	Marks Distribution			Duration of End Semester Examination
L	T	P	C	Internal Assessment	End Semester Examination	Total	
4	0	0	4	Maximum Marks: 40	Maximum Marks: 60	100	3 Hours
				Minimum Marks: 16	Minimum Marks: 24	40	

Guidelines for setting Question Paper: Question paper of end semester examination will be of 60 marks. The question paper will consist of five sections A, B, C, D, E. Sections A,B,C and D will have 2 questions of 12 marks each and Section E has short answer type questions consisting of six parts of 02 marks each. The candidates will attempt five questions in all, i.e. one question each from sections A,B,C, D and the compulsory question from section E. In the question paper, the questions available in sections A,B,C and D will be covered from Unit-I, Unit-II, Unit-III and Unit-IV respectively and Section-E will covered the whole syllabus.

Course Objectives (COs)

- The main objective of this course is to introduce the basic theory underlying the different components and phases of a compiler like parsing,code generation etc..

Unit I	08 Lectures
Introduction to compilers and interpreters – Overview of compilation and different phases, Issues in compilation – structure of a compiler – compiler writing tools – bootstrapping – notations and concepts for languages and grammars – regular expressions – context free grammar, derivations and parse trees, BNF notations.	
Unit II	08 Lectures
Context of a lexical analyzer: Construction of lexical analyzer, deterministic and non-deterministic finite automata. Syntax analyzer, context free grammars, top down parsing, brute force parser, recursive descent parser, LL(1) parser, Bottom up parsing, operator precedence parsing, simple precedence parsing, LR parser, LALR parser, YACC – the parser generator.	
Unit III	07 Lectures
Compile time error handling: Error detection, reporting, recovery and repair. Intermediate languages, postfix notation, syntax trees, parse trees, three address code, triples and indirect triples. Translation of assignment statements, Boolean expressions, Syntax Directed Definition, S-attributed, L-attributed, translation Scheme, Applications of Syntax directed translation.	
Unit IV	07 Lectures
Run time storage management: Storage allocation and referencing data in block structured language, storage allocation. Code optimization, sources of optimization, loop optimization, DAG and optimization of basic blocks. Code generation, a machine model, next use information register allocation and assignment, a simple code generator, code generation from DAG's, Peephole optimization.	

Course Learning Outcomes (CLOs)

- The student will be able to use formal grammars to specify the syntax of language and understand key issues of ambiguity and associativity.

Suggested Readings:

- Alfred V. Aho, Jeffrey D. Ullman (2001), Principles of compiler design, Indian student edition, Pearson Education, New Delhi, India.

MCA-7305(ii) Data Warehouse and Mining							
Teaching Scheme			Credit	Marks Distribution			Duration of End Semester Examination
L	T	P		Internal Assessment	End Semester Examination	Total	
4	0	0	4	Maximum Marks: 40	Maximum Marks: 60	100	3 Hours
				Minimum Marks: 16	Minimum Marks: 24	40	

Guidelines for setting Question Paper: Question paper of end semester examination will be of 60 marks. The question paper will consist of five sections A, B, C, D, E. Sections A,B,C and D will have 2 questions of 12 marks each and Section E has short answer type questions consisting of six parts of 02 marks each. The candidates will attempt five questions in all, i.e. one question each from sections A,B,C,D and the compulsory question from section E. In the question paper, the questions available in sections A,B,C and D will be covered from Unit-I, Unit-II, Unit-III and Unit-IV respectively and Section-E will cover the whole syllabus.

Course Objectives (COs)

- Interpret the problem in real time applications, and apply the appropriate data mining technique for effective results.

Unit I	08 Lectures
Introduction: Fundamentals of data mining, Data Mining Functionalities, Classification of Data Mining systems, Major issues in Data Mining, Data Warehouse and OLAP Technology for Data Mining Data Warehouse, Multidimensional Data Model, Data Warehouse Architecture, Data Warehouse Implementation, Further Development of Data Cube Technology, From Data Warehousing to Data Mining.	
Unit II	08 Lectures
Data Preprocessing: Needs Preprocessing the Data, Data Cleaning, Data Integration and Transformation, Data Reduction, Discrimination and Concept Hierarchy Generation, Online Data Storage. Data Mining Primitives, Languages, and System Architectures: Data Mining Primitives, Data Mining Query Languages.	
Unit III	07 Lectures
Concepts Description: Characterization and Comparison: Data Generalization and Summarization-Based Characterization, Analytical Characterization: Analysis of Attribute Relevance, Mining Class Comparisons: Discriminating between Different Classes, Mining Descriptive Statistical Measures in Large Databases.	
Unit IV	07 Lectures
Classification and Prediction-1: Issues Regarding Classification and Prediction, Classification by Decision Tree Induction, Bayesian Classification, Rule-Based Classification, Classification by Back propagation. Classification and Prediction-2: Support Vector Machines, Association Classification, Other Classification Methods, Prediction, Accuracy and Error Measures, Evaluating the Accuracy of a Classifier or Predictor	

Course Learning Outcomes (CLOs)

- The student will be able to describe concepts of data mining and its functionalities.
- Explain the primitives and query languages of data mining

Suggested Readings

- Pang- Ning Tan, Michael Steinbach, Vipin Kumar, "Introduction to Data Mining", 1st Edition, Pearson Education, 2012.

MCA-7305(iii) Ethical Hacking							
Teaching Scheme			Credit	Marks Distribution			Duration of End Semester Examination
L	T	P		Internal Assessment	End Semester Examination	Total	
4	0	0	4	Maximum Marks: 40	Maximum Marks: 60	100	3 Hours
				Minimum Marks: 16	Minimum Marks: 24	40	

Guidelines for setting Question Paper: Question paper of end semester examination will be of 60 marks. The question paper will consist of five sections A, B, C, D, E. Sections A,B,C and D will have 2 questions of 12 marks each and Section E has short answer type questions consisting of six parts of 02 marks each. The candidates will attempt five questions in all, i.e. one question each from sections A,B,C,D and the compulsory question from section E. In the question paper, the questions available in sections A,B,C and D will be covered from Unit-I, Unit-II, Unit-III and Unit-IV respectively and Section-E will cover the whole syllabus.

Course Objectives (COs)

- The main objective of this course is explaining the objectives of ethical hacking and understanding various password attacks. Discuss the fundamental ideas of database attacks.

Unit I	08 Lectures
Ethical Hacking Overview: Hacking Life Cycle, Legal Issues in Ethical Hacking, Hacking Terminology, Gathering Facts, CP/IP Concepts Review, Network and Computer Attacks. Network Enumeration and Foot Printing: DNS Query, WHOIS Query, OS Finger Printing, Banner Grabbing, CERT-In Guideline: CERT-In Guideline for Securing Wireless Access Points/Routers, Credit Card, Email, Web Server Security, Auditing and Logging, Securing Home Computers, SQL Server Security.	
Unit II	08 Lectures
Programming For Security Professionals: Web Application Vulnerabilities, Buffer Overflow Attack, Session Hijacking. Code Injection Attacks: Cross Site Scripting (CSS) Attack, SQL injection Attack. Required Lab goals: Basics, Information Required to Attack Target, Finding Critical Bugs in Servers.	
Unit III	07 Lectures
Types of Hacking : Password Hacking, Windows Hacking, Logging by Pass, Network Hacking, and Anonymity and Email Hacking. Web Servers Hacking, Session Hijacking, Surveillance, Desktop, Server and OS Vulnerabilities Required Lab. Goals: Methods of Password Encryption and Decryption learn to remain anonymous over the Internet.	
Unit IV	07 Lectures
Database Attacks, Hacking Wireless Networks, Cryptography, Network Protection Systems, Trojan and Backdoor Applications, Legal Resources, Virtualization and Ethical Hacking. Required Lab Goals: ways to maintain access to a system using Trojan and backdoor, attacking database server and wireless networks	

Course Learning Outcomes (CLOs)

- The students will be able to understand basic ethical hacking terminology.
- Ability to identify type of attack. Ability to understand the ways to secure password and databases.

Suggested Readings

- Stuart McClure Joel Scambray, Solutions, George Kurtz -Hacking Exposed— Network Security Secrets & solutions.
- Ethical Hacking and Network Defence, Michael T. Simpson, Kent Backman, James Corley.

MCA-7305(iv) Machine Learning							
Teaching Scheme			Credit	Marks Distribution			Duration of End Semester Examination
L	T	P		Internal Assessment	End Semester Examination	Total	
4	0	0	4	Maximum Marks: 40	Maximum Marks: 60	100	3 Hours
				Minimum Marks: 16	Minimum Marks: 24	40	

Guidelines for setting Question Paper: Question paper of end semester examination will be of 60 marks. The question paper will consist of five sections A, B, C, D, E. Sections A,B,C and D will have 2 questions of 12 marks each and Section E has short answer type questions consisting of six parts of 02 marks each. The candidates will attempt five questions in all, i.e. one question each from sections A,B,C,D and the compulsory question from section E. In the question paper, the questions available in sections A,B,C and D will be covered from Unit-I, Unit-II, Unit-III and Unit-IV respectively and Section-E will cover the whole syllabus.

Course Objectives (COs)

- This course will serve as a comprehensive introduction to various topics in machine learning. At the end of the course, the students should be able to design and implement machine learning solutions to classification, regression, and clustering problems; and be able to evaluate and interpret the results of the algorithms.

Unit I	08 Lectures
Introduction: History of Machine Learning, Programs Vs learning algorithms, Machine Learning definition, types of learning, hypothesis space and inductive bias, evaluation, cross-validation, Instance based learning, Feature reduction, Collaborative filtering-based recommendation, Gradient Descent learning.	
Unit II	08 Lectures
Supervised Learning: General notions - Bayes optimality, curse of dimensionality, overfitting and model, selection, bias vs. variance tradeoff, generative vs. discriminative for parameter estimation, feature selection, and etc.	
Unit III	07 Lectures
Unsupervised Learning: Clustering and density estimations - Kmeans/vector quantization, mixture models, Dimensionality reduction - linear and nonlinear methods, Principal components analysis.	
Unit IV	07 Lectures
Deductive Learning: Probability theory and Bayes rule. Naive Bayes learning algorithm. Parameter smoothing. Generative vs. discriminative training. Logistic regression. Bayes nets and Markov nets for representing dependencies.	

Course Learning Outcomes (CLOs)

- The students will be able to understand the fundamental concepts in machine learning and popular machine learning algorithms.

Suggested Readings

- Introduction to Machine learning, Nils J.Nilsson .
- Machine learning for dummies, IBM Limited ed, by Judith Hurwitz and Daniel Kirsch.

MCA-7305(v) Introduction to Quantum Computing							
Teaching Scheme			Credit	Marks Distribution			Duration of End Semester Examination
L	T	P		Internal Assessment	End Semester Examination	Total	
4	0	0	4	Maximum Marks: 40	Maximum Marks: 60	100	3 Hours
				Minimum Marks: 16	Minimum Marks: 24	40	

Guidelines for setting Question Paper: Question paper of end semester examination will be of 60 marks. The question paper will consist of five sections A, B, C, D, E. Sections A,B,C and D will have 2 questions of 12 marks each and Section E has short answer type questions consisting of six parts of 02 marks each. The candidates will attempt five questions in all, i.e. one question each from sections A,B,C,D and the compulsory question from section E. In the question paper, the questions available in sections A,B,C and D will be covered from Unit-I, Unit-II, Unit-III and Unit-IV respectively and Section-E will covered the whole syllabus.

Course Objectives (COs)

- The main objective of this course is basic introduction to quantum mechanics, linear algebra and familiarity with the Dirac notation is provided first to get one's quantum moorings right. This is then followed by an introductory treatment of quantum computation and quantum information covering aspects of quantum entanglement, quantum algorithms, quantum channels. Rudimentary quantum computing is introduced using the IBM quantum computer and associated simulators.

Unit I	08 Lectures
Introduction: Elementary quantum mechanics, linear algebra for quantum mechanics, Quantum states in Hilbert space, The Bloch sphere, Density operators, generalized measurements, no-cloning theorem.	
Unit II	08 Lectures
Quantum correlations: Bell inequalities and entanglement, Schmidt decomposition, super dense coding, teleportation. Quantum cryptography: Quantum key distribution.	
Unit III	07 Lectures
Quantum gates and algorithms: Universal set of gates, quantum circuits, Solovay-Kitaev theorem, Deutsch-Jozsa algorithm, factoring.	
Unit IV	07 Lectures
Programming a quantum computer: The IBMQ, coding a quantum computer using a simulator to carry out basic quantum measurement and state analysis.	

Course Learning Outcomes (CLOs)

- The students will be able to provide a self-contained, comprehensive introduction to quantum computing, focusing on the design and analysis of quantum algorithms, as well as covering topics in quantum information and quantum cryptography, such as: quantum teleportation, quantum money, and post-quantum cryptography.

Suggested Readings

- Quantum Computation and Quantum Information, M. A. Nielsen & I. Chuang, Cambridge University Press.
- Quantum Computing, A Gentle Introduction, Eleanor G. Rieffel and Wolfgang H. Polak MIT press (2014).

MCA-7306(i) Internet of Things(IoT)							
Teaching Scheme			Credit	Marks Distribution			Duration of End Semester Examination
L	T	P		Internal Assessment	End Semester Examination	Total	
4	0	0	4	Maximum Marks: 40	Maximum Marks: 60	100	3 Hours
				Minimum Marks: 16	Minimum Marks: 24	40	

Guidelines for setting Question Paper: Question paper of end semester examination will be of 60 marks. The question paper will consist of five sections A, B, C, D, E. Sections A,B,C and D will have 2 questions of 12 marks each and Section E has short answer type questions consisting of six parts of 02 marks each. The candidates will attempt five questions in all, i.e. one question each from sections A,B,C, D and the compulsory question from section E. In the question paper, the questions available in sections A,B,C and D will be covered from Unit-I, Unit-II, Unit-III and Unit-IV respectively and Section-E will covered the whole syllabus.

Course Objectives (COs)

- To introduce the terminology, technology and its applications.
- To introduce the implementation of web-based services on IoT devices.

Unit I	08 Lectures
Introduction: Definition and Characteristics of IoT: Sensors, Actuators, Challenges of IoT, Advantages and Disadvantages of IoT, How IoT Works? Application of IoT: IoT Smart Home Applications, Wearable Applications, Smart City Applications, Smart Grids Applications, Industrial Internet Applications, Connected Health (Digital Health) Applications, Smart Retail Applications, Transportation Application, Education Applications, Government Applications, Smart Agriculture Applications .	
Unit II	08 Lectures
IoT Architecture and Protocols: Introduction, Functional Blocks of IoT, IoT Architecture, IoT Protocol Physical Design of IoT: What is IoT Design, Industry 4.0 Design Principles, IoT Design Tools, Physical Design of IoT. Logical Design of IoT: IoT Functional Blocks, Communication Models, IoT Communication APIs.	
Unit III	07 Lectures
Understanding Basics of Arduino IDE: Features, Types of Arduino, Exploring Arduino Uno, Installation and Understanding Basics of Arduino IDE, Structure of Arduino Program, Collecting and Showing Data through Serial Port. Basics of C Language using Arduino IDE: Loops and Control Statement, Function in C using Arduino.	
Unit IV	07 Lectures
Practical Using Arduino Interfacing Sensors: Interfacing Light Emitting Diode (LED) - Blinking Led, Interfacing Button and Led-Led Blinking When Button is Pressed, Interfacing Temperature Sensor (LM35), Interfacing Temperature & Humidity Sensor (e.g. DHT11), Interfacing Liquid Crystal Display (LCD) Display Data Generated by Sensor on LCD.	

Course Learning Outcomes (CLOs)

- The student will be able to understand the IoT value chain structure (device, data cloud), application areas and technologies involved. Understand IoT sensors and technological challenges faced by IoT devices.

Suggested Readings:

- Pawan Thakur, Susheela Pathania, “Internet of Things (IoT)”, Satya Prakashan, New Delhi.
- Peter Friess,'Internet of Things – From Research and Innovation to Market Deployment', River Publishers, 2014.

MCA-7306(ii) Soft Computing							
Teaching Scheme			Credit	Marks Distribution			Duration of End Semester Examination
L	T	P		Internal Assessment	End Semester Examination	Total	
4	0	0	4	Maximum Marks: 40	Maximum Marks: 60	100	3 Hours
				Minimum Marks: 16	Minimum Marks: 24	40	

Guidelines for setting Question Paper: Question paper of end semester examination will be of 60 marks. The question paper will consist of five sections A, B, C, D, E. Sections A,B,C and D will have 2 questions of 12 marks each and Section E has short answer type questions consisting of six parts of 02 marks each. The candidates will attempt five questions in all, i.e. one question each from sections A,B,C, D and the compulsory question from section E. In the question paper, the questions available in sections A,B,C and D will be covered from Unit-I, Unit-II, Unit-III and Unit-IV respectively and Section-E will covered the whole syllabus.

Course Objectives (COs):

- Introduce students to soft computing concepts and techniques and foster their abilities in designing and implementing soft computing based solutions for real world and engineering problems.

Unit I	08 Lectures
Introduction: Neural Networks, Application Scope of Neural Network, Fuzzy Logic, Genetic Algorithm, Hybrid Systems and Soft computing. Artificial Neural Network: Fundamental Concept, Evolution of Neural Networks, Basic Models of Artificial Neural Network, Important Terminologies of ANNs, McCulloch-Pitts Neuron, and Hebb Network.	
Unit II	08 Lectures
Introduction to Fuzzy Logic, Classical Sets and Fuzzy Sets: Introduction to Fuzzy logic, Classical Sets (Crisp Sets), Operations of Classical Sets, Fuzzy Sets Operations. Classical Relations and Fuzzy Relations: Cartesian Product of Relation, Classical Relation, Fuzzy Relations, Tolerance and Equivalence Relations, No interactive Fuzzy Sets. Membership Functions: Features of Membership Functions, Fuzzification and Defuzzification.	
Unit III	07 Lectures
Fuzzy Rule Base and Approximate Reasoning: Introduction, Truth Values and Table in Fuzzy Logic, Fuzzy Propositions, Fuzzy Reasoning, Fuzzy Inference System. Fuzzy Decision Making: Individual Decision Making, Multiperson Decision Making, Multiobjective Decision Making, Multiattribute Decision Making, Fuzzy Bayesian Decision Making.	
Unit IV	07 Lectures
Hybrid Soft Computing Techniques: Neuro-Fuzzy Hybrid Systems, Generic Neuro-Hybrid Systems, Genetic Fuzzy Hybrid and Fuzzy Genetic Hybrid Systems, . Applications of Soft Computing: A Fusion Approach of Multispectral Images with SAR (Synthetic Aperture Rader), Optimization of Traveling Salesmen Problem using Genetic Algorithm Approach, Genetic Algorithm Based Internet Search Technique, Soft computing Based Hybrid Fuzzy Controllers.	

Course Learning Outcomes (CLOs)

- The student will be able to understand soft computing techniques and their role in problem solving.

Suggested Readings:

- Principles of Soft Computing by S.N. Sivanandam, S.N. Deepa, Wiley.
- Fuzzy Logic and Soft Computing by Bernadette Bouchon-Meunier, World Scientific.

MCA-7306(iii) Computer Vision							
Teaching Scheme			Credit	Marks Distribution			Duration of End Semester Examination
L	T	P		Internal Assessment	End Semester Examination	Total	
4	0	0	4	Maximum Marks: 40	Maximum Marks: 60	100	3 Hours
				Minimum Marks: 16	Minimum Marks: 24	40	

Guidelines for setting Question Paper: Question paper of end semester examination will be of 60 marks. The question paper will consist of five sections A, B, C, D, E. Sections A,B,C and D will have 2 questions of 12 marks each and Section E has short answer type questions consisting of six parts of 02 marks each. The candidates will attempt five questions in all, i.e. one question each from sections A,B,C,D and the compulsory question from section E. In the question paper, the questions available in sections A,B,C and D will be covered from Unit-I, Unit-II, Unit-III and Unit-IV respectively and Section-E will cover the whole syllabus.

Course Objectives (COs)

- The objectives of this course are to develop the understanding of the basic principles and techniques of image processing and image understanding, and to develop your skills in the design and implementation of computer vision software.

Unit I	08 Lectures
Digital image representation, image acquisition, storage and processing. fundamental steps in image processing, Introduction to Image Processing, Computer Vision and Computer Graphics, Monocular imaging system, basics of Image Formation, Radiance, Irradiance, BRDF, color etc.	
Unit II	08 Lectures
Introduction, definition, Active vision system, Machine vision components, hardware's and algorithms, segmentation, data reduction, feature extraction, edge detection, image recognition and decisions, identification, resolution passive and active stereo imaging.	
Unit III	07 Lectures
Structure of the human eye, image formation, brightness adaptation and discrimination, a simple image model, uniform and non-uniform sampling and quantization, distance measures, Fourier and wavelet descriptors, Multiresolution analysis, Hough transforms and other simple object recognition methods, PCA, HMM and GMM.	
Unit IV	07 Lectures
Face detection and Face recognition, Eigen faces, Active appearance and 3D shape models of faces. Surveillance – foreground-background separation, particle filters, Chamfer matching, tracking, and occlusion, combining views from multiple cameras, human gait analysis.	

Course Learning Outcomes (CLOs)

- The students will be able to describe image representation and basics of computer vision.
- Develop various applications using computer vision.

Suggested Readings

- Davies, E. R., Computer & Machine Vision, 4th Edition, Academic Press, 2012.
- Mark, N.A.S., Feature Extraction & Image Processing for Computer Vision, 3rd Edition, Academic Press, 2012.

MCA-7306(iv) Introduction to Blockchain							
Teaching Scheme			Credit	Marks Distribution			Duration of End Semester Examination
L	T	P	C	Internal Assessment	End Semester Examination	Total	
4	0	0	4	Maximum Marks: 40	Maximum Marks: 60	100	3 Hours
				Minimum Marks: 16	Minimum Marks: 24	40	

Guidelines for setting Question Paper: Question paper of end semester examination will be of 60 marks. The question paper will consist of five sections A, B, C, D, E. Sections A,B,C and D will have 2 questions of 12 marks each and Section E has short answer type questions consisting of six parts of 02 marks each. The candidates will attempt five questions in all, i.e. one question each from sections A,B,C, D and the compulsory question from section E. In the question paper, the questions available in sections A,B,C and D will be covered from Unit-I, Unit-II, Unit-III and Unit-IV respectively and Section-E will covered the whole syllabus.

Course Objectives (COs)

- To allow the students to explore the driving force behind the crypto-currency Bitcoin. Along with the Decentralization, Cryptography, Bitcoins with its alternative coins.

Unit I	08 Lectures
Introduction to Block chain, Block chain Technology Mechanisms & Networks, Block chain Origins, Objective of Block chain, Block chain Challenges, Transactions and Blocks.	
Unit II	08 Lectures
P2P Systems, Keys as Identity, Digital Signatures, Hashing, and public key cryptosystems, private vs. public Blockchain.CAP theorem and block chain, Block chain Network, Mining Mechanism,	
Unit III	07 Lectures
Distributed Consensus, Gas Limit, Transactions and Fee, Anonymity, Reward, Chain Policy, Life of Block chain application, Soft & Hard Fork. Benefits and limitations of block chain.	
Unit IV	07 Lectures
Nakamoto consensus, Proof of Work, Proof of Stake, Proof of Burn, Difficulty Level, Sybil Attack, Energy utilization and alternate. Decentralization using block chain, Methods of decentralization, Routes to decentralization, Decentralized organizations.	

Course Learning Outcomes (CLOs)

- The student will be able to Understand the types, benefits and limitation of block chain.
- Explore the block chain decentralization and cryptography concepts.
- Enumerate the Bitcoin features and its alternative options.

Suggested Readings:

- Mastering Block chain - Distributed ledgers, decentralization and smart contracts explained, Author - Imran Bashir, Packt Publishing Ltd, Second Edition, ISBN 978-1- 78712-544-5, 2017.

MCA-7306(v) Software Engineering							
Teaching Scheme			Credit	Marks Distribution			Duration of End Semester Examination
L	T	P		Internal Assessment	End Semester Examination	Total	
4	0	0	4	Maximum Marks: 40	Maximum Marks: 60	100	3 Hours
				Minimum Marks: 16	Minimum Marks: 24	40	

Guidelines for setting Question Paper: Question paper of end semester examination will be of 60 marks. The question paper will consist of five sections A, B, C, D, E. Sections A,B,C and D will have 2 questions of 12 marks each and Section E has short answer type questions consisting of six parts of 02 marks each. The candidates will attempt five questions in all, i.e. one question each from sections A,B,C, D and the compulsory question from section E. In the question paper, the questions available in sections A,B,C and D will be covered from Unit-I, Unit-II, Unit-III and Unit-IV respectively and Section-E will covered the whole syllabus.

Course Objectives (COs):

- Basic knowledge and understanding of the analysis and design of complex systems.
- Ability to apply software engineering principles and techniques.

Unit I	08 Lectures
Introduction: Evolving Role of Software, Software Engineering, Changing nature of Software, Software Myths, Terminologies, Role of management in software development Software Process and desired Characteristics, Software Life Cycle Models: Build & Fix Model, Water Fall Model, Incremental Process Model, Evolutionary Process Models, Unified Process, Comparison of Models, Other Software Processes, Selection of a Model, Software Requirements Analysis & Specifications.	
Unit II	08 Lectures
Agile Methodology: Agile Modeling, Its use and advantages, Scrum, Advantages and disadvantages of agile Modeling. Software Architecture: It's Role, Views, Component & Connector View and its architecture style, Architecture Vs Design, Deployment View & Performance Analysis, Documentation, Evaluation. Software Project Planning: Size estimation, Cost Estimation, COCOMO, COCOMO – II.	
Unit III	07 Lectures
Function Oriented Design: Design principles, Module level Concepts, Notation & Specification, Structured Design Methodology, Verification. Object-Oriented Design: OO Analysis & Design, OO Concepts, Design Concepts, Noun Phrase Analysis, Sequence & Collaboration Diagram, CRC cards, UML – Class Diagram, Other diagrams & Capabilities, Design Methodology – Dynamic and Functional Modeling, Internal Classes & Operations.	
Unit IV	07 Lectures
Software Metrics: Token Count, Data Structure Metrics, Information Flow Metrics, Object-Oriented Metrics, Use Case Oriented Metrics, Web Engineering Project Metrics, Metric Analysis. Software Maintenance & Certification: Maintenance, Maintenance Process and Models, Estimation of Maintenance Costs, Regression Testing, Reverse Engineering, Software Re engineering, Configuration Management, Documentation, Requirements of Certification, Types.	

Course Learning Outcomes (CLOs)

- Ability to understand software lifecycle development models.
- Ability to understand and apply software requirements engineering techniques.
- Ability to understand and apply software testing techniques.

Suggested Readings:

- Pankaj Jalote, —An Integrated Approach to Software Engineering, Narosa Publishing House.
- K.K. Aggrawal and Yogesh Singh, —Software Engineering, New Age International (P) Ltd.

MCA- 7307P Lab VII: Analysis and Design of Algorithm Lab							
Teaching Scheme			Credit	Marks Distribution			Duration of End Semester Examination
L	T	P	C	Internal Assessment	End Semester Examination	Total	
0	0	4	2	Maximum Marks: 40	Maximum Marks: 60	100	3 Hours
				Minimum Marks: 16	Minimum Marks: 24	40	

Course Objectives (COs)

- The objectives of the Machine Learning Lab course are to introduce students to the basic concepts and techniques of Machine Learning.

Total Lab Hours: 40 (Max)

Suggested List of Practical

Topics:

- To implement Linear search and Binary search and analyze its time complexity.
- To implement Huffman Coding and analyze its time complexity.
- To implement Dijkstra's Algorithm.
- Implement N-Queen problem using Backtracking.
- Design and implement program to find minimum spanning tree using Kruskal's algorithm.
- Implement various String Matching algorithms.
- Write a program to find solution for knapsack problem using greedy method.
- To implement Bellman Ford algorithm and analyze its complexity.

Course Learning Outcomes (CLOs)

- The student will be able to review and apply some common Machine Learning algorithms.
- Perform experiments in Machine Learning using real-world data.

MCA- 7308P Lab VIII: Mobile Application Deployment Lab							
Teaching Scheme			Credit	Marks Distribution			Duration of End Semester Examination
L	T	P	C	Internal Assessment	End Semester Examination	Total	
0	0	4	2	Maximum Marks: 40	Maximum Marks: 60	100	3 Hours
				Minimum Marks: 16	Minimum Marks: 24	40	

Course Objectives (COs)

- The objective of Mobile Application Deployment lab is to understand the components and structure of mobile application development frameworks for android .

Total Lab Hours: 40 (Max)

Suggested List of Practical

Topics:

- Develop an application that uses GUI components, Font and Colors .
- Develop an application that uses Layout Managers and Event Listeners.
- Write an application that draws basic Graphical Primitives on the screen.
- Develop an application that make use of databases.
- Develop an application that make use of Notification Manager.
- Implement an application that uses Multi-threading.
- Develop a native application that uses GPS location Information.
- Develop a mobile application to send an E-mail.

Course Learning Outcomes (CLOs)

The students will be able to develop mobile application using GUI, Layouts, Event Listeners and Databases.

MCA- 7309P Lab IX: Innovative Project Identification							
Teaching Scheme			Credit	Marks Distribution			Duration of End Semester Examination
L	T	P	C	Internal Assessment	End Semester Examination	Total	
0	0	4	2	Maximum Marks: 40	Maximum Marks: 60	100	
				Minimum Marks: 16	Minimum Marks: 24	40	3 Hours

Course Objectives (COs)

- To implement various languages on different platforms.

Total Lab Hours: 40 (Max)

Instructions:

Project must be the part of practical report. Student has to develop a real life application by using Programming languages like Java, Web Technology, Android, Python etc. The students should show their own work, and must not be copied from any other resources.

Course Learning Outcomes (CLOs)

- The students will be able to develop applications to solve real world problems using different languages.

HIMACHAL PRADESH TECHNICAL UNIVERSITY HAMIRPUR



Syllabus

for

MCA

(SEMESTER-IV)

Amended as per NEP-2020

(w.e.f. the Academic Year 2023-2024)

Semester-IV						
Course Code	Course Type	Subject Title/ Subject Name	Credits	Marks		
MCA-7401(A)	CC	Innovative Project Design & Implementation (Industrial / In-House)	16	Min	Max	
		(A). Synopsis			20	50
		(B). Innovative Mid Project Report			20	50
		(C). Innovative Project Seminars(02)			20	50
		(D). Innovative Project Progress Report(02)			20	50
		(E). Innovative Project Report			40	100
		(F). Major Project Viva-Voce			40	100
Total				160	400	
Or						
MCA-7401(B)	CC	Research (Industrial / In-House)	16	Marks		
					Min	Max
		(A). Synopsis			20	50
		(B). Mid Research Report			20	50
		(C). Research Work Seminars (02)			20	50
		(D). Research Work Progress Report(02)			20	50
		(E). Research Work Report			40	100
(F). Research Work Viva-Voce			40	100		
		Total		160	400	

Instructions:

1. For Student: Students should take this project work very seriously, as these efforts will be considered as 4 to 6 months experience in most of the software companies or research. Topics selected should be complex and large enough to justify as a MCA Innovative project or Research. The student can undertake the topics/specifications from the MCA- 7309P Lab IX: Innovative Project Identification. The project or research should be genuine and original in nature and should not be copied from anywhere else. If found copied the project report will be forwarded to the exam discipline committee of the University as an unfair means case for necessary action.

2. For Internal and External Examiner: Marks distribution for Industrial Training and Project Work

Internal Examiner 200 Marks will be awarded by internal examiner on the bases of Synopsis, Innovative Mid Project or Research Report, Innovative Project or Research Seminars(2) and Innovative Project or Research Progress Report(2).

External Examiner 200 Marks will be awarded by external examiner on the bases of Innovative Project or Research Report and Vive-Voice. The Project or Research Report (100 Marks), End Semester External Vive-Voice (100 Marks).

**GUIDELINES FOR
INNOVATIVE PROJECT DESIGN & IMPLEMENTATION(INDUSTRIAL / IN-HOUSE)
OR
RESEARCH (INDUSTRIAL / IN-HOUSE)**

1.1. Introduction

The Master of Computer Applications (MCA) programme prepares the students to take up positions as Systems Analysts, Systems Designers, Software Engineers, Programmers, Research and Project Managers in any field related to information technology. We had therefore imparted you the comprehensive knowledge covering the skills and core areas of computer science courses with equal emphasis on the theory and practical. The MCA students are encouraged to spend at least 4 to 6 months working on a project or research work preferably in a software industry or any research organization.

1.2. Objective

The objective of the MCA project work is to develop quality software solution. During the development of the project you should involve in all the stages of the software development life cycle like requirements engineering, systems analysis, systems design, software development, testing strategies and documentation with an overall emphasis on the development of reliable software systems. The primary emphasis of the project work is to understand and gain the knowledge of the principles of software engineering practices so as to participate and manage a large software engineering projects in future. After the completion of this project or research work the student should be able to:

1. Describe the SDLC, Evaluate systems requirements, problem definition, determine how to collect information to determine requirements
2. Evaluate feasibility and Operational feasibility for the project.
3. Work on data collection methods for fact finding.
4. Construct and evaluate data flow diagrams, data dictionaries, process, alternative tools for the analysis process.
5. Create and evaluate such alternative graphical tools as systems flow charts and state transition diagrams.
6. Decide the S/W requirement specifications and H/W requirement specifications.
7. Plan the systems design phase of the SDLC.
8. Distinguish between logical and physical design requirements.
9. Design and evaluate system, evaluate systems inputs, evaluate validity checks for input data.
10. Design and evaluate user interfaces for input, file structures and Estimate storage requirements.
11. Construct and evaluate (ER) diagrams for RDBMS related projects.
12. Decide the various processing systems to include distributed, client/server, online and others. Perform project cost estimates using various techniques.
13. Schedule projects using both GANTT and PERT charts. Perform coding for the project.
14. Documentation requirements and prepare and evaluate systems documentation.
15. Perform various systems testing techniques/strategies to include the phases of testing.
16. Systems implementation and its key problems.
17. Generate various reports

18. Brief the maintenance procedures and the role of configuration management in operations.
19. To decide the future scope and further enhancement of the system.
20. Plan for several appendices to be placed in support with the project report documentation.
21. Work effectively as an individual or as a team member to produce correct, efficient, well organized and documented programs in a reasonable time.
22. Recognize problems that are amenable to computer solutions, and knowledge of the tools necessary for solving such problems.
23. Develop of the ability to assess the implications of work performed.
24. Get good exposure and command in one or more application areas and on the software
25. Develop quality software using the software engineering principles.
26. Develop of the ability to communicate effectively.

1.3. Mode and Type of the Innovative Project & Research

The majority of the students are expected to work on a real-life project preferably in some industry/ Research and Development Laboratories/Educational Institution/Software Company. The student can also formulate a project problem with the help of her/his Guide and submit the project proposal of the same in the college within 20 days at the starting of Innovative Project & Research work. Approval of the project or Research proposal is mandatory which will be evaluated by internal examiner appointed by respective college Principal or Director. If approved, the student can commence working on it and complete it. Use the latest versions of the software packages for the development of the project.

1.4. Eligibility criteria of a Project Guide

A person having Ph.D. / M.Tech. in Computer Science with a minimum of one year of experience.

Or

A person having MCA / M.Sc. (CS/IT) with minimum 2 years experience, preferably in software development.

1.5. System Development Project & Research

In fourth semester, the Chairman/Head of the Department will assign a guide/supervisor, to each candidate for his/her project and Research work. The candidate shall be required to maintain his/her project diary (logbook) of work in the organization. Each student will be required to give at least two seminars on his/her project work. Each student is required to submit three copies of his/her project reports in the Department after completion of the project work which will be evaluated by external examiner.

1.6. Arrangement of Contents of Project Report

The sequence in which the project report material should be arranged and bound should be as follows:

1. Title page (Annexure I)
2. Title page (Annexure II)
3. Certificate of Originality (Annexure III)
4. Industry/Company Certificate (Annexure IV)
5. Project-Team Certificate (Only if applicable) (Annexure V)
6. Acknowledgement (Annexure VI)

7. Innovative Project Design & Implementation: Table of Contents (Annexure VII)
8. Research Report : Table of Contents((Annexure VII)

1.7. Number of Copies to be submitted

Students should submit three copies to the Head of the Department on or before the specified date along with the soft copy of project report and executable file of application software properly write in CR, entitled “Title of the Project Report”, “Name” and “Roll No” of the candidate with black or blue permanent marker. The Head of the Department should send

1. One copy to the University. (After final viva-voice)
2. One copy to the Department library. (After final viva-voce)
3. One copy to the Internal Examiner (Before final viva-voce) and
4. One copy to the student concerned (Not to be Submit to the Head of the Department).

1.8. Size of Project and Research Report

The size of project report should not be less than 100 pages and should not exceed 150 pages of typed matter reckoned from the first page of INTRODUCTION to the last page.

1.9. Arrangement of Contents of Project and Research Report

The sequence in which the project report material should be arranged and bound should be as follows:

1. Title page (Annexure I)
2. Title page (Annexure II)
3. Certificate of Originality (Annexure III)
4. Industry/Company Certificate (Annexure IV)
5. Project-Team Certificate (Only if applicable) (Annexure V)
6. Acknowledgement (Annexure VI)
7. Table of Contents (Annexure VII)
8. Project Report pages as per table of contents.

1.10 Page Dimensions and Margin

The dimensions of the final bound copies of the project report should be 290mm x 205mm. Standard A4 size (297mm x 210mm) paper may be used for preparing the copies.

The final five copies of the project report (at the time of submission) should have the following page margins:

Top edge	:	30 to 35 mm
Bottom edge	:	25 to 30 mm
Left side	:	35 to 40 mm
Right side	:	20 to 25 mm

The project report should be prepared on good quality white paper preferably not lower than 80 grams /Sq. Meter. Tables and figures should conform to the margin specifications. Large size figures should be photographically or otherwise reduced to the appropriate size before insertion.

1.11. Typing Instructions

The impressions on the typed copies should be black in color. The font used for entire project report should be Times New Roman with following sizes and styles.

Text Type	Size	Style	Illustration
Main headings	16	Bold/ UPPERCASE	1. Introduction
Division headings	14	Bold/ Capitalized Each Word	2.4. Feasibility Analysis
Sub-division headings	12	Bold/ Capitalized Each Word	2.4.1. Technical Feasibility
Normal text	12	Normal / Sentence case	Any text under Division headings or Sub-division headings
Any highlighted word under normal text	12	Bold/ Sentence case	Any highlighted text under Division headings or Subdivision headings
Bibliography and References	10	Normal/Sentence case	Bibliography and reference items.

Uniformity in the font of letters in the same project report shall be observed. A sub-heading at the bottom of a page must have at least two full lines below it or else it should be carried over to the next page. The last word of any page should not be split using a hyphen. One and a half spacing should be used for typing the general text. Single spacing should be used for typing:

1. Long Tables
2. Long quotations
3. Foot notes
4. Multiline captions
5. References

All quotations exceeding one line should be typed in an indented space – the indentation being 15mm from either margins. Double spacing should be used for typing all the Certificates and Acknowledgement. Don't alter the format mentioned under Annexure I, II, III, V and VII. The text for Annexure IV and VI will be decided by the Industry/Company, Candidate respectively.

TITLE OF PROJECT OR RESEARCH REPORT

By

Name of Student

A PROJECT OR RESEARCH REPORT

Submitted to the

HIMACHAL PRADESH TECHNICAL UNIVERSITY (HPTU)

HAMIRPUR, H.P.



*In partial fulfillment of the requirements for the award of the
degree*

of

MASTER OF COMPUTER APPLICATIONS (MCA)

IN

4th Semester Training

<College Logo>

Month, Year

Name of the Department

College/Institution Name

Annexure-II (Format for inside Cover Page)

PROJECT REPORT

ON

TITLE OF PROJECT OR RESEARCH

**SUBMITTED IN PARTIAL FULFILLMENT FOR AWARD OF
DEGREE**

In

**MASTER OF COMPUTER APPLICATION
(BATCH)**

BY

NAME OF THE STUDENT

ROLL NO

UNDER THE GUIDANCE OF

NAME OF GUIDE (Internal, External)

College Logo

Name of the Department

College/ Institution Name

Month, Year

**HIMACHAL PRADESH TECHNICAL UNIVERSITY,
HAMIRPUR-H.P.**

Annexure-III

CERTIFICATE OF ORIGINALITY

I hereby declare that the Project entitled “**Title of the Innovative Project or Research**” submitted to the Department of Computer Applications, **Name of the College** in partial fulfillment for the award of the Degree of **MASTER OF COMPUTER APPLICATIONS** in session <**Session**> in an authentic record of my own work carried out under the guidance of Dr./Ms./Mr. “**External and Internal Guide Name**” and that the Project has not previously formed the basis for the award of any other degree.

Place:

(Signature of the candidate)

Date:

Name of Student

(Roll No.)

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

(Signature of Internal Guide)

(Signature of External Guide)

Name

Name

Designation

Designation

Annexure-IV

INDUSTRY/COMPANY CERTIFICATE

Industry/Company Certificate on the letter head of respective company duly signed by External Guide/Project Manager/Supervisor.

Date:

(Name of External Guide/Project or Research Manager/Supervisor)

Designation:

Name of Organization:

Address:

Annexure-V

PROJECT-TEAM CERTIFICATE

(In case the entire project is developed by team not by an individual student)

Name of the Project:

Team Members Details:

Sr. No.	Name of Candidate	College/Institute Name	University /Organization
1.			
2.			
3.			
4.			

I hereby declared that the project entitled “**Title of the Project or Research**”, is developed by the above mentioned team. In the team “**Name of candidate**” has developed/designed the following module(s) under the esteemed guidance of “**Name of External Guide with his/her Designation**”.

1. Details of module one including its name and candidate’s responsibilities.
2. Details of module two including its name and candidate’s responsibilities.
3.
4.

Date:

(Name of Project Manager/Supervisor)

Designation:

Name of Organization:

Address:

Annexure-VI

ACKNOWLEDGEMENT

Acknowledgement should be brief and should not exceed one page when typed double spacing.

Date:

Signature of Candidate

Name:

Roll No.:

TABLE OF CONTENTS

S.No.	Contents and Chapter Name	Page No.	
		From	To
1	CHAPTER 1: INTRODUCTION	1-10	
	1.1. Introduction	1	3
	1.2. Objective	3	4
	1.3. About Organization	4	10
2	CHAPTER 2 : SYSTEM ANALYSIS	11-46	
	2.1. Identification of Need (Problem Specification, Existing System, Proposed System, Feasibility Analysis)	11	25
	2.2. Project Planning and Project Scheduling (PERT Chart and Gantt Chart)	26	32
	2.3. Software requirement specifications (SRS)	33	36
	2.4. Software Engineering	37	40
	2.5. Data models (DFD, Control Flow diagrams, State Diagrams/Sequence diagrams, Entity Relationship Model, Class Diagrams/CRC Models/Collaboration Diagrams/Use-case Diagrams/Activity Diagrams and other models depending upon your project requirements)	40	46
3	CHAPTER 3: SYSTEM DESIGN		
	3.1. Modularization details		
	3.2. Data integrity and constraints		
	3.3. Database design, Procedural Design/Object Oriented Design		
	3.4. User Interface Design		
	3.5. Test Cases (Unit Test Cases and System Test Cases)		
4	CHAPTER 4: CODING		
	4.1 SQL commands/Object description for (i) data or database creation (along with constraints), (ii) data collection, cleaning and generation and insertion into data structure or tables and (iii) access rights for different users		
	4.2. Complete Project Coding		
	4.3. Comments and Description of Coding segments		
	4.4. Standardization of the coding (Code Efficiency, Error handling, Parameters calling/passing, Validation checks)		
5	CHAPTER 5: TESTING		
	5.1 Testing techniques and Testing strategies used		
	5.2. Testing Plan used		
	5.3. Test reports for Unit Test Cases and System Test Cases		
5	CHAPTER 6: SYSTEM SECURITY MEASURES & REPORTS		
	6.1. Database/data Security		
	6.2. Creation of User profiles and access rights		
	6.3. Reports (sample layouts should be placed)		
7	CHAPTER 7: CONCLUSION AND FURTHER ENHANCEMENT		
	7.1. Conclusion		
	7.2 Further Scope and Enhancement		
	Bibliography		
	Appendices (if any)		
	Glossary		

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

NOMENCLATURE

ACRONYMS (if any)

Chapter 1: Introduction

1.1

1.2 :

1.6 Organization of the Thesis

Chapter 2: Review of Past Work and Problem Formulation

Chapter 3:

Chapter 4: : : : :

Chapter (Second Last) should focus on Results and Discussion/Analysis

Chapter (Last): Conclusions and Scope for Future Work

APPENDIX-A (if any) : :

REFERENCES